Conducting polymers

A polymer which can conduct electricity is termed as conducting polymers.

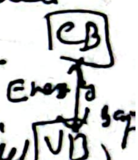
eg: poly aniline, poly ~~pyrrole~~ pyrrole etc.

Classification →

- Conducting polymers
  - Intrinsically conducting
    - Conducting polymers have conjugated π e's
    - Dopped conducting polymers
  - Extrinsically conducting
    - Conductive element filled polymers
    - Blended conducting polymers

Conducting Polymers have conjugated π e⁻ns

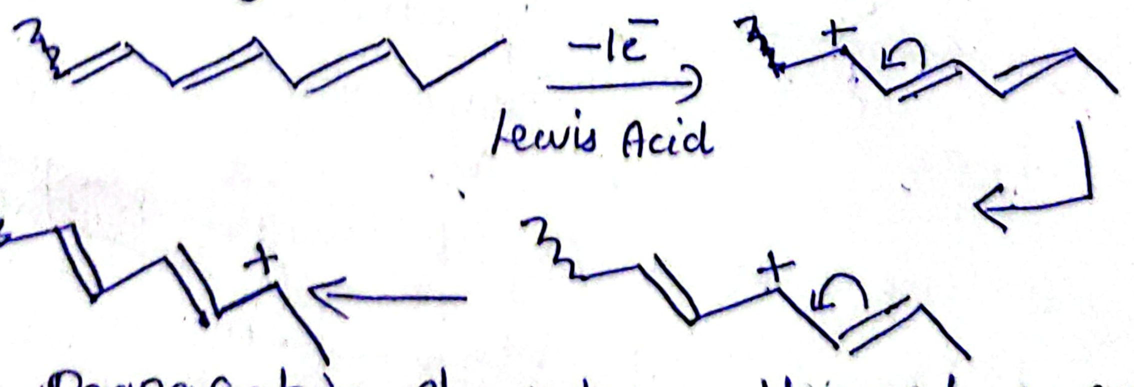
Here the polymers have conjugated double bonds along the back bone of the polymer. The orbitals of conjugated π electrons overlap over the entire backbone of the polymer, resulting in the formation of valence band and conduction band. Electrical conduction ~~wd~~ would occur when electron from valance band are excited to conduction band either thermally ~~thera~~ or Photolytically. Since VB and CB separated by a significant band gap, conductivity of polymers are not high.



Doped conducting polymers

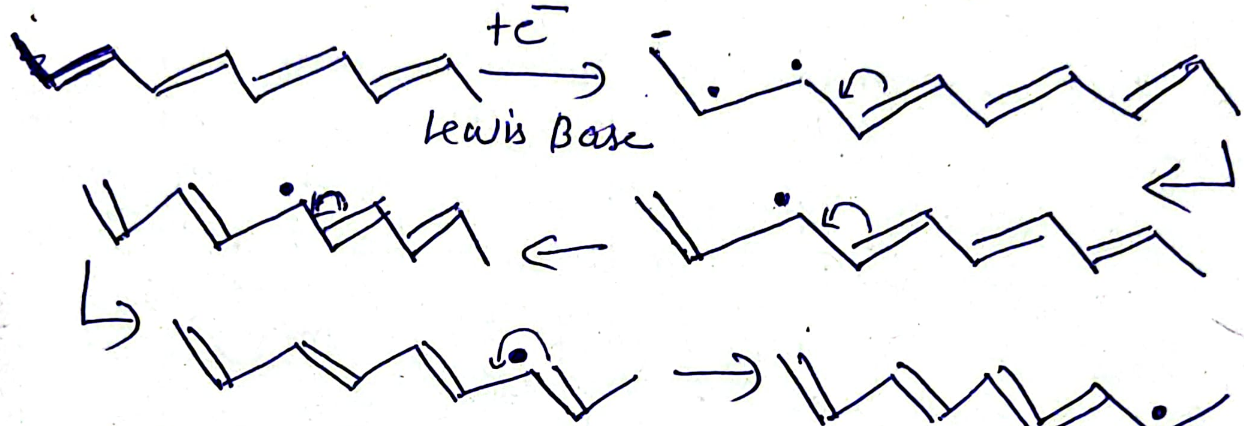
Conductivity of a polymer can be increased by creating positive or negative charge on polymer backbone by oxidation or reduction. This process is termed as doping.

P- doping: It is done by oxidation process. The polymer is treated with Lewis acid. Then holes [(+) charge) are created and can move along the polymer chain



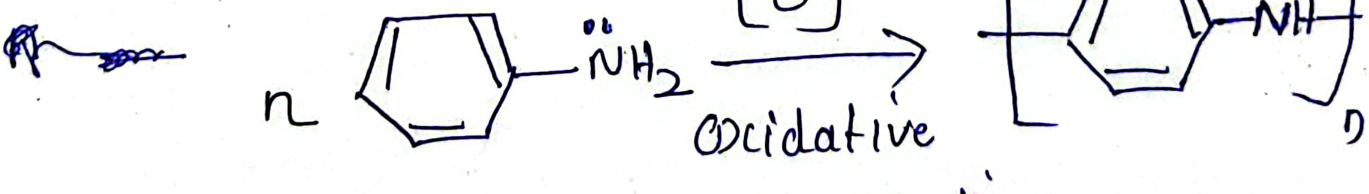
Propagation of polaron through a conjugated polymer chain by shifting of double bonds.

n-doping - is done by reduction process. Here electrons are introduced into the Polymer chain having conjugated double bonds. The reduction of polyacetylene by a Lewis base leads to the formation of Polaron.



Poly aniline

Prepared by Electrochemical method. It is done by dipping Platinum electrodes in aqueous solution of aniline and a potential of 0.7V is applied, polyaniline is formed on the surface of positive electrode



Oxidative polymerisation $+ nH_2O$

Properties

- Polyaniline is a light weight material
- It possess very high chemical resistance.
- Doped polyaniline is a good conductor comparable to copper

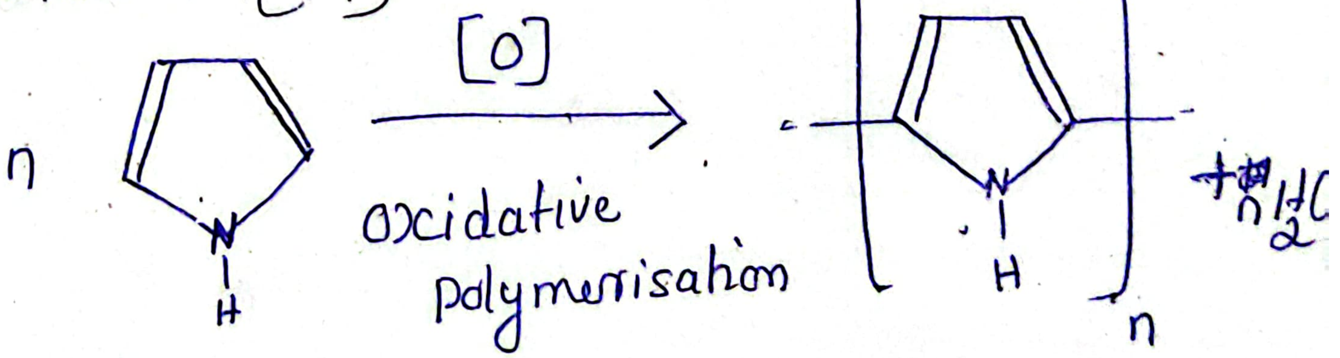
Polyaniline shows electrochromism

Applications

- Use as Electrode material for rechargeable batteries
- used in biosensors and solar energy absorption
- used in printer circuit board
- used to make light weight rechargeable batteries

Poly pyrrole

Poly pyrrole can be prepared by chemical oxidative polymerisation of pyrrole using ammonium per sulfate (sulphate) as oxidant [1:1]



(chemical oxidative polymerisation can be carried out in a beaker by mixing 0.1 m aq. solution of pyrrole and 0.1 m ammonium persulphate)

Properties

- Poly Pyrrole has excellent electrical, thermal and Mechanical Properties.
- It has good environmental stability and higher conductivity
- Poly pyrrole film are yellow but darken in air due to oxidation.
- It has very high chemical resistance.
- They are amorphous and give weak diffraction

Applications

- Widely used in bio sensors and immunosensors.
- used in smart windows and display.
- used in light weight rechargeable batteries and electronic devices.
- It have applications in tissue engineering
- used in analytical sensors, Photovoltaic devices.
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